Lawrence S. Giron Jr.

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EDUCATION

University of Alaska Anchorage - College of Engineering — *Anchorage, AK*
Master of Science, Mechanical Engineering — *Spring 2024 – Spring 2025*
- Overall GPA: 4.00/4.00
- NASA EPSCoR Graduate Funded Research Thesis: Atmospheric Corrosion Studies of Metal Alloys
- Graduate Courses: Unmanned Aircraft Systems Operations, Advanced Fluid Mechanics, Advanced Materials Engineering, Automatic Control, Engineering Finite Element Analysis

University of Alaska Anchorage - College of Engineering — *Anchorage, AK*
Bachelor of Science, Mechanical Engineering and Minor in Mathematics — *Fall 2020 - Spring 2023*
- Overall GPA: 3.82/4.00
- Mechanical Engineering Departmental Honors
- Capstone Project: Thermal Analysis of Materials and Design for Manufacturing
- Advanced Engineering Courses: Manufacturing Design, Corrosion Processes and Engineering, Mechanical Vibrations, Aerodynamics

EXPERIENCE

University of Alaska Anchorage - Graduate Research Assistantship — *Anchorage, AK*
NASA EPSCoR Funded Graduate Research Thesis: Atmospheric Corrosion Studies of Metal Alloys — *Spring 2024 – Summer 2025*
- Conducted post-exposure analysis of metal alloys through ASTM and ISO standards
- Performed Scanning Electron Microscopy (SEM) and Electron-dispersive X-ray Spectroscopy (EDXA) analysis of metal alloys, Surface Morphology Profilometry, and Cyclic Corrosion Test Chamber corrosion testing
- Explored the correlation of corrosion rate of metal alloys to weather and exposure conditions
- Developed a dose-response model to improve corrosion rate estimation in subarctic regions

NASA Glenn Research Center - Intelligent Control and Autonomy Intern — *Cleveland, OH*
National Aeronautics and Space Administration - Aeronautics Research Mission Directorate — *Fall 2023*
- Expanded the Subsonic Single Aft Engine (SUSAN) hybrid-electric passenger aircraft concept powertrain power and propulsion model for the optimization and management of thermal health
- Thermal Modeling and Analysis of electrical components of the aircraft simulation model with additions made to the MATLAB Simulink Electrical Modeling and Thermal Analysis Toolbox (EMTAT) developed by NASA (EMTAT v1.3.3 LEW-20097-1)

NASA Langley Research Center - Research Associate — *Hampton, VA*
National Aeronautics and Space Administration - Aeronautics Research Directorate Academy — *Summer 2023*
- Research involved under the In-Time System-Wide Safety Assurance
- Designed and tested a cost-effective disaster mitigation tool to provide an aerial perimeter point of view for first responders in natural disaster situations
- Studied, designed, and tested parachute recovery systems to provide stable and steady descents of launched sensors

University of Alaska Anchorage - Research Assistant — *Anchorage, AK*
Magnetostrictive Sensor Research Project Assistant — *Spring 2023*
- Designed and tested a temperature compensated magnetostrictive corrosion detection sensor using Fiber Bragg Grating sensors to record the thermal response of alloys
- Tracked magnetic flux leaks through induced mechanical strain with fiber optic strain sensors to detect pipe wall thinning due to metallurgical degradation and decay

University of Alaska Anchorage - Teaching Assistant — *Anchorage, AK*
Summer Engineering Academy Teaching Assistant — *Summer 2022*
- Assisted in teaching engineering principles and practices to students grade 6-12 that included: Tetrix Robotics, Civil Engineering Structures, Wing Aerodynamics, and Corrosion Chemistry

CERTIFICATIONS & SKILLS & MEMBERSHIPS

- **AMPP – Association for Materials Protection and Performance:** AMPP Membership
- **Design and Modeling Software:** MATLAB, Simulink, ANSYS, SOLIDWORKS, Autodesk Fusion 360 and AutoCad
- **Machines:** Hitachi Scanning Electron Microscopy (SEM) and Oxford Electron-dispersive X-ray Spectroscopy (EDXA), NANOVEA Profilometry, Q-LAB Q-FOG Cyclic Corrosion Test Chamber

CONFERENCES, PROJECT DISPLAYS, & SYMPOSIUMS

- **Energy & Mobility Conference & Expo**
- **NASA Aviation Day 2023**
  - Displayed development in ongoing projects at NASA GRC
- **NASA Glenn Research Center Space Apps Competition**
  - Developed the Virtual Utility for Locating, Containing, and Assisting Notification (VULCAN Fire Response Ops) designed to enhance fire prevention, monitoring, and response operations
    - First Place Award
- **National Space Grant Council of Space Grant Directors Meeting**
- **Alaska Space Grant Program Education and Research Symposium**
  - Poster – "NASA Langley Research Center Academy & NASA Glenn Research Center Internship"
- **ConocoPhillips Alaska Arctic Science and Engineering Endowment Award Symposium**
  - Poster – "Characterizing Environmental Severity and Atmospheric Corrosivity Categorization in Alaska"
- **AMPP – The Association for Materials Protection and Performance Annual Conference + Expo 2025**
  - Paper – Raghu Srinivasan, Lawrence Giron Jr., and Jozef Hunter, "Characterizing Environmental Severity and Atmospheric Corrosivity Categorization in Alaska", Paper No. C2025-00566, 2025 AMPP, March 2025, Nashville, TN.
  - Student Poster Session – "The Effects of Exposure Angle on Atmospheric Corrosivity Categorization in Alaska"
- **Student Research & Creative Scholarship Showcase**
  - Poster – "Characterizing Environmental Severity and Atmospheric Corrosivity Categorization in Alaska"
    - Graduate Students Division Excellence in Research Award

PAPERS

- K. Anderson, C. Brander, E. Carrasco, R. Courville, A. de la Guardia, J. Fensler, **L. Giron Jr.**, J. Oberlies, W. Pankiewicz, and G. Wachter, "Compact Lightweight Aerial Sensor System (CLASSy)", Contractor or Grantee Report No. 20230011971, NASA Langley Research Center, Aug. 11, 2023. [Online]. Available: https://ntrs.nasa.gov/citations/20230011971
- Raghu Srinivasan, **Lawrence Giron Jr.**, and Jozef Hunter, "Characterizing Environmental Severity and Atmospheric Corrosivity Categorization in Alaska", Paper No. C2025-00566, 2025 AMPP, March 2025, Nashville, TN. Available: https://doi.org/10.5006/C2025-00566
- **L. Giron Jr.**, "Characterizing Environmental Severity and Atmospheric Corrosivity Categorization in Alaska." Order No. 31996628, University of Alaska Anchorage, United States -- Alaska, 2025. Available: https://www.proquest.com/dissertations-theses/characterizing-environmental-severity-atmospheric/docview/3195353162/se-2
- (In-Progress) **Lawrence Giron Jr.,** Raghu Srinivasan, Jozef Hunter, Matthew Cullin; Atmospheric Corrosion Modeling and Corrosivity Categorization of 1008 Carbon Steel in Alaska. *CORROSION*.